Lecture 17: Taylor and Maclaurin Series

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand and work out the series representation of a function.
- ☐ Define Taylor and Maclaurin series and identify the conditions under which they represent a function.
- ☐ Derive the Taylor and Maclaurin series for a given function using derivatives at a point.
- ☐ Compute Taylor polynomials of specified degrees to approximate functions near a point.
- ☐ Expand elementary functions using Maclaurin series.

Series Representation

In last lecture, we have seen how to generate a power series for certain function by using the geometric series. But now if we are given any function, can we always represent it with a power series?

Taylor series - a power series generated by a function that are infinitely differentiable.

Within the interval of convergence I, a power series is a continuous function with derivative of all orders. How about the other way around?

Assume that $f(x)$ is the sum of a power series about $x = a$,

$$f(x) = \sum_{n=0}^{\infty} a_n (x-a)^n \qquad f(a) = a_0$$

$$= a_0 + a_1(x-a) + a_2(x-a)^2 + \cdots + a_n(x-a)^n + \cdots$$

with a positive radius of convergence.

Series Representation

By performing term-by-term differentiation repeatedly,

$$f'(x) = a_1 + 2a_2(x-a) + 3a_3(x-a)^2 + \dots + na_n(x-a)^{n-1} + \dots$$

$$f'(a) = a_1$$

$$f''(x) = 2a_2 + 2\cdot 3a_3(x-a) + \dots + n(n-1)a_n(x-a)^{n-2} + \dots$$

$$f''(a) = 2a_2$$

$$\vdots$$

$$f^{(n)}(x) = 2\cdot 3\cdot 4\cdot \dots \cdot n\,a_n + \text{some infinite polynomial with factor } (x-a) + \dots$$

$$f^{(n)}(a) = n!\,a_n$$

So now, $f(a) = c_0 \Rightarrow c_0 = \frac{f(a)}{1} = \frac{f(a)}{0!}$ ← we define $0! = 1$

$f'(a) = c_1 \Rightarrow c_1 = \frac{f'(a)}{1!}$

$f''(a) = 2c_2 \Rightarrow c_2 = \frac{f''(a)}{2!}$

$f'''(a) = 2 \cdot 3\, c_3 = 3!\, c_3 \Rightarrow c_3 = \frac{f'''(a)}{3!}$

$\vdots$ $\qquad\qquad$ $\vdots$

$f^{(n)}(a) = n!\, c_n \Rightarrow c_n = \frac{f^{(n)}(a)}{n!}$

within the radius of convergence.

So, if $f(x)$ has a power series representation, then

$$f(x) = \frac{f(a)}{0!} + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \dots$$

$$\boxed{f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n}$$ ← Taylor series of f. at $x = a$.

Taylor and Maclaurin Series

Let f be a function with derivatives of all orders throughout some interval containing a as an interior point. Then the Taylor series generated by f at $x = a$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(x)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots .$$

The Maclaurin series of f is the Taylor series generated by f at $x = 0$, or

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots .$$

Taylor and Maclaurin Series

Find the Taylor series generated by $f(x) = \frac{1}{x}$ at $a = 2$. Where, if anywhere, does the series converge to $\frac{1}{x}$?

Taylor series at $x=2$: $\sum_{n=0}^{\infty} \frac{f^{(n)}(2)}{n!}(x-2)^n$

$f(x) = \frac{1}{x}$

$f'(x) = -\frac{1}{x^2}$

$f''(x) = \frac{1\cdot 2}{x^3}$

$f'''(x) = -\frac{1\cdot 2\cdot 3}{x^4}$

$f^{(4)}(x) = \frac{1\cdot 2\cdot 3\cdot 4}{x^5}$

$\vdots$

$f^{(n)}(x) = (-1)^n \frac{n!}{x^{n+1}} \Rightarrow f^{(n)}(2) = (-1)^n \frac{n!}{2^{n+1}}$

So the Taylor series of $f(x) = \frac{1}{x}$ at $x = 2$ is

$$\sum_{n=0}^{\infty} \frac{(-1)^n \cancel{n!}}{2^{n+1}} (x-2)^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} (x-2)^n$$

$$\cancel{n!}$$

$$= \frac{1}{2} - \frac{1}{4}(x-2) + \frac{1}{8}(x-2)^2 - \frac{1}{16}(x-2)^3 + \dots$$

This is a geometric series with

$a = \frac{1}{2}$, $r = -\frac{1}{2}(x-2)$,

so it converges to $\frac{1}{x}$ when $|r| < 1$

$$\Rightarrow \left|-\tfrac{1}{2}(x-2)\right| < 1$$

$$\Rightarrow \left|\tfrac{1}{2}(x-2)\right| < 1$$

$$\Rightarrow 0 < x < 4$$

Taylor Polynomials

Let f be a function with derivatives of order k for $k = 1, 2, \ldots, N$ in some interval containing a as an interior point. Then, for any integer n from 0 through N, the Taylor polynomial of order n generated by f at $x = a$ is the polynomial

← first n terms of the Taylor Series.

$$P_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(k)}(a)}{k!}(x-a)^k + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

Note: We speak of a Taylor polynomial of order n rather than degree n because $f^{(n)}(a)$ may be zero.

Eg: $f(x) = \cos x$ at $x = 0$

$P_0(x) = f(0) = 1$

$P_1(x) = f(0) + f'(0)(x-0) = 1 + (-\sin 0)(x-0) = 1$

← Taylor polynomial of order 1 but it is a polynomial of degree 0.

Taylor Polynomials

(Madaurin Series)

Example: Find the Taylor series and the Taylor polynomials generated by $f(x) = e^x$ at $x = 0$.

Taylor Series of $x=0$: $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$?

$f(x) = e^x$

$f'(x) = e^x$

$\vdots$

$f^{(n)}(x) = e^x$ for all $n \geq 0 \Rightarrow f^{(n)}(0) = e^0 = 1$ for all $n \geq 0$

So, $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$

$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

Taylor Polynomial $P_n(x) = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}$

Taylor Polynomials

Example: Find the Taylor series and the Taylor polynomials generated by $f(x) = \cos x$ at $x = 0$.

$f(x) = \cos x$ $\quad$ $f'(x) = -\sin x$

$f''(x) = -\cos x$ $\quad$ $f'''(x) = \sin x$

$f^{(4)}(x) = \cos x$ $\quad$ $f^{(5)}(x) = -\sin x$

even order of derivative

odd order of derivative.

$f^{(2n)}(x) = (-1)^n \cos x$ $\quad$ $f^{(2n+1)}(x) = (-1)^{n+1} \sin x$

At $x=0$, $f^{(2n)}(x) = (-1)^n$ $\quad$ $f^{(2n+1)}(x) = 0$

So $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$

$$= 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

Taylor Polynomial $P_{2n}(x) = P_{2n+1}(x) = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots + \frac{(-1)^n x^{2n}}{(2n)!}$

Exercise: Find the Taylor series for $\sin x$ at $x = 0$.

Taylor Polynomials

Example: Expand $f(x) = \ln(1+x)$ at $x = 0$.

$f(x) = \ln(1+x)$ $\qquad f(0) = 0$

$f'(x) = \frac{1}{1+x}\frac{d}{dx}(1+x) = \frac{1}{1+x}$

$f''(x) = -\frac{1}{(1+x)^2}$

$f'''(x) = \frac{2}{(1+x)^3}$

$f^{(4)}(x) = -\frac{2\cdot 3}{(1+x)^4}$

$\vdots$

$f^{(n)}(x) = (-1)^{n+1}\frac{(n-1)!}{(1+x)^n}$

$f^{(n)}(0) = (-1)^{n+1}(n-1)!$ $\qquad n \geq 1$

So the Taylor Series of $f(x) = \ln(1+x)$ at $x=0$

is $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = 0 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(n-1)!}{n!\ n} x^n$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

Taylor Polynomials

Example: Compute the Taylor polynomials of order 7 generated by $f(x) = \sinh x$ at $x = 0$.

Exercise.

$$P_7(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!}$$

Taylor Polynomials

Consider

$$f(x) = \begin{cases} 0, & x = 0 \\ e^{-1/x^2}, & x \neq 0 \end{cases}$$

At $x=0$, $f(x)=0$

$f'(x)=0$

$\vdots$

$f^{(n)}(x)=0$ for all n.

So, the Taylor Series generated by f at $x=0$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = 0$$

Notice that this Taylor series converges to 0 for all x but $f(x) \neq 0$ for all $x \neq 0$

The Taylor series generated by f at $x=0$ does not converges to the function itself.

Extra:

Arithmetic operation on Taylor Series are valid on the intersection of their interval of convergence.

Eg: $\frac{1}{3}(2x + x\cos x) = \frac{2}{3}x + \frac{x}{3}\cos x$

$= \frac{2}{3}x + \frac{x}{3}\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots\right)$

$= \frac{2x}{3} + \frac{x}{3} - \frac{x^3}{3\cdot 2!} + \frac{x^5}{3\cdot 4!} - \frac{x^7}{3\cdot 6!} + \dots$

$= x - \frac{x^3}{6} + \frac{x^5}{3\cdot 4!} - \frac{x^7}{3\cdot 6!} + \dots$

Ex: $e^x \cos x = 1 + x - \frac{1}{3}x^3 - \frac{1}{6}x^4 + \dots$

$(\underbrace{\quad}_{e^x})(\underbrace{\quad}_{\cos x})$

Furthermore, we can use the Taylor Series of the function f to find the Taylor series of $f(u(x))$ where $u(x)$ is a continuous function.

Eg: $\cos(2x)$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

Since $u(x) = 2x$ is a cts function.

$$\cos(2x) = \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n}}{(2n)!}$$

replace x with $2x$ of the Taylor series of $\cos x$.

$$= 1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + \cdots$$